

NASA Nearest Earth Objects (1910–2024) Hazard Classification Using Machine Learning

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1. OBJECTIVES

Near-Earth Objects (NEOs) are asteroids and comets whose orbits bring them into proximity with Earth's path around the Sun. While the vast majority pose no immediate threat, a small fraction are classified as Potentially Hazardous Asteroids (PHAs). This is a designation with meaningful implications for planetary defence planning, emergency preparedness, and the long-term safety of human civilization.

The central objective of this project is to build a machine learning classification system that can identify which observable asteroid properties are most strongly associated with a hazardous orbit designation. Using the NASA Nearest Earth Objects dataset (1910–2024), containing 338,199 observations, the project addresses the following core question:

Can observable physical and orbital properties of near-Earth asteroids (such as diameter, velocity, miss distance, and brightness) reliably predict whether an object will be classified as hazardous?

An important framing note must be acknowledged upfront. NASA's official Potentially Hazardous Asteroid classification is determined by two hard thresholds: a Minimum Orbit Intersection Distance (MOID) of less than 0.05 astronomical units, and an absolute magnitude brighter than 22. Neither of these precise threshold values is present in the dataset as a standalone feature. Rather than treating this as a blocker, the project frames the model as a triage tool; one that learns which combinations of observable properties correlate most strongly with hazardous designation, rather than claiming to reverse-engineer NASA's classification formula directly.

This framing is both scientifically honest and practically useful. A model trained to surface the highest-risk candidates from large observation catalogues has direct operational value in prioritizing which asteroids warrant deeper investigation, even if it does not replace the authoritative NASA classification process.

The secondary objectives of the project are to compare the performance of multiple classifier architectures under class imbalance, to engineer physics-motivated features that add predictive power beyond the raw observational columns, and to evaluate model outputs through metrics that reflect the asymmetric cost of false negatives in a safety-critical domain.

2. DATA PREPARATION

2.1 Data Source

The dataset was sourced from Kaggle: NASA Nearest Earth Objects (1910–2024), compiled by user ivansher from NASA's publicly available NEO observation records. The dataset is available at:

<https://www.kaggle.com/datasets/ivansher/nasa-nearest-earth-objects-1910-2024>

The raw dataset contains 338,199 rows and 9 columns, representing individual close-approach observations of near-Earth asteroids spanning over a century of records. Each row corresponds to a single observation event, meaning the same asteroid may appear multiple times across different flyby events.

2.2 Dataset Features

Feature	Type	Description
neo_id	Integer	Unique NASA identifier for each asteroid
name	String	Asteroid designation (e.g., 162117 (1998 SD15))
absolute_magnitude	Float	Intrinsic brightness (lower = brighter/larger)
estimated_diameter_min	Float	Minimum estimated diameter (km)
estimated_diameter_max	Float	Maximum estimated diameter (km)
orbiting_body	String	Always 'Earth' in this dataset
relative_velocity	Float	Velocity relative to Earth (km/h)
miss_distance	Float	Closest approach distance (km)
is_hazardous	Boolean	Target variable: NASA hazard classification

Figure 1: A table with the names of all the available columns in the raw dataset, the column data type, and a short description of the column.

2.3 Data Quality

Overall data quality was high. A systematic check for missing values revealed that only `absolute_magnitude`, `estimated_diameter_min`, and `estimated_diameter_max` contained nulls; all at the same 0.008% rate (28 rows). This near-negligible rate is consistent with a small number of poorly-observed objects where magnitude data was unavailable. Given the trivial proportion relative to 338,000+ observations, these rows were dropped rather than imputed, avoiding the risk of introducing synthetic signals near class boundaries.

No duplicate rows were identified. All numeric columns were confirmed to be correctly typed as float64 or int64. The `orbiting_body` column contained only 'Earth' throughout the dataset and was therefore dropped as it provides no discriminative information. The `neo_id` and `name` columns were similarly removed as identifier fields with no predictive value.

2.4 Class Imbalance

The target variable exhibited a significant class imbalance: 295,009 non-hazardous objects (87.2%) versus 43,162 hazardous objects (12.8%). This 87/13 split is an important consideration for model design and evaluation. A naive model that always predicts non-hazardous would achieve 87.2% accuracy; a figure that is meaningless from a safety perspective. All subsequent modelling decisions account for this imbalance, either through class weighting parameters or through evaluation metrics that do not reward majority-class dominance.

2.5 Feature Engineering

Exploratory data analysis revealed that `estimated_diameter_min` and `estimated_diameter_max` have a perfect correlation of 1.00, making them redundant. Rather than retaining both, they were consolidated into a single `estimated_diameter_mean` feature, reducing dimensionality without losing information.

Three additional physics-motivated features were engineered to increase predictive signal:

- **Kinetic Energy Proxy**
 - $\text{kinetic_energy} = \text{estimated_diameter_mean}^2 \times \text{relative_velocity}^2$. This approximates the destructive potential of an impact event, combining size and speed into a single risk-relevant quantity.
- **Size-to-Distance Ratio**
 - $\text{size_to_distance_ratio} = \text{estimated_diameter_mean} / \text{miss_distance}$. This captures how physically large an object is relative to how close it passes, providing a proximity-adjusted size signal.
- **Magnitude-Velocity Interaction**
 - $\text{magnitude_velocity} = \text{absolute_magnitude} \times \text{relative_velocity}$. This captures the interaction between intrinsic brightness (a proxy for size) and approach speed.

After feature engineering, the final dataset contained 338,171 observations and 7 features (excluding the target). All features were standardized using `StandardScaler` prior to modelling, and subsequently normalized using `MinMaxScaler` for models sensitive to feature scale.

2.6 Tools and Libraries

All data preparation, analysis, and modelling was conducted in Python using the following libraries: `pandas` and `NumPy` for data manipulation; `matplotlib` and `seaborn` for visualization; `scikit-learn` for preprocessing, modelling, and evaluation; `XGBoost` and `LightGBM` for gradient boosting classifiers. The analysis was conducted in a Jupyter notebook environment with `KaggleHub` used for programmatic dataset download.

3. MODEL DESIGN

3.1 Exploratory Analysis and PCA

Prior to modelling, a Principal Component Analysis (PCA) was conducted to assess the separability of the two classes in a reduced feature space. With log-transformed and standardized features, the first two principal components captured 78.6% of total variance (PC1: 54.4%, PC2: 24.2%). The PCA loadings revealed that PC1 is primarily a size dimension (high diameter loading, low absolute magnitude loading) while PC2 captures orbital dynamics (miss distance and velocity).

The 2D PCA projection showed partial but meaningful class separation: hazardous asteroids concentrate in the central region of the feature space while non-hazardous objects dominate the right side and extremes. The significant overlap between classes confirmed that perfect classification is not achievable with this feature set, a finding consistent with the absence of MOID from the dataset, and appropriately calibrated expectations for model performance.

Correlation analysis identified `absolute_magnitude` as the strongest individual predictor of hazard status ($r = -0.34$), followed by `relative_velocity` ($r = 0.19$) and `estimated_diameter_mean` ($r = 0.16$).

Miss_distance showed a near-zero correlation ($r = -0.007$), suggesting that how close an asteroid passes on a given observation is not, by itself, a reliable predictor of its hazard classification.

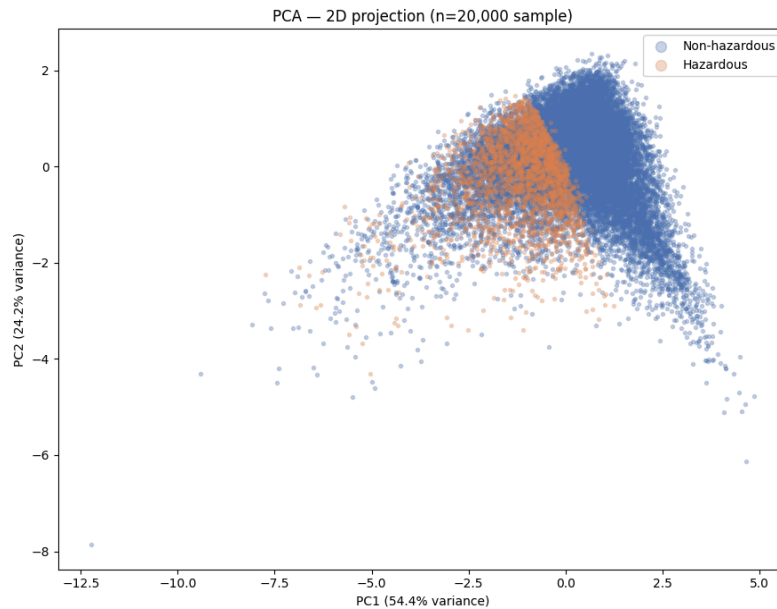


Figure 2: A 2D projection of the top two principal components of the dataset

3.2 Baseline Model

A DummyClassifier using the most_frequent strategy was established as a performance floor. This baseline always predicts non-hazardous and achieves 87.36% accuracy; the exact proportion of the majority class. Every trained model must surpass this baseline on hazardous-class metrics to demonstrate genuine learning.

3.3 Logistic Regression

Logistic Regression was selected as the first substantive model due to its interpretability and its established role as a standard binary classification benchmark. It models the log-odds of hazard classification as a linear function of the feature vector. The model was trained with max_iter=1000 to ensure convergence on the 270,000+ training observations. No class weighting was applied in the initial run, which proved consequential, the model effectively defaulted to the majority class, mirroring the dummy baseline on hazardous-class recall.

3.4 Random Forest

A Random Forest with 100 decision trees was trained using class_weight='balanced', which adjusts sample weights inversely proportional to class frequency. This was the first model to apply explicit class balancing. Random Forest was selected for its robustness to feature scale, its native ability to capture non-linear interactions, and its built-in feature importance ranking; useful for understanding which engineered features contributed most to classification.

3.5 XGBoost

XGBoost (eXtreme Gradient Boosting) was included as a high-performance gradient boosting baseline. Unlike Random Forest, which builds trees independently, XGBoost builds trees sequentially where each tree corrects the residual errors of its predecessor. The model was trained with 100 estimators at default hyperparameters, without explicit class weighting, allowing a direct comparison against the class-balanced Random Forest.

3.6 AdaBoost

AdaBoost was tested as an adaptive boosting alternative, using decision stumps (`max_depth=1`) as weak learners with a learning rate of 0.1 and 100 estimators. AdaBoost iteratively reweights misclassified samples to focus subsequent estimators on harder cases. However, without class weighting, the model collapsed to the majority-class baseline, producing identical outputs to the DummyClassifier.

3.7 Support Vector Machine (Linear and RBF)

Two SVM variants were tested. A Linear SVM (LinearSVC) was trained with `C=100` and `class_weight='balanced'`. The regularization parameter `C` was varied across `{1, 10, 100}` with virtually identical results, indicating the data is not linearly separable; a conclusion supported by the PCA overlap analysis. An RBF kernel SVM was then tested to allow non-linear decision boundaries. Due to the $O(n^2)$ computational complexity of kernel SVMs, training was performed on a 20,000-observation stratified subsample with `C=100` and `gamma='scale'`.

3.8 LightGBM

LightGBM (Light Gradient Boosting Machine) was the final model tested. LightGBM uses a leaf-wise tree growth strategy rather than the level-wise approach of XGBoost, allowing it to achieve higher accuracy with faster training times on large datasets. The model was configured with 100 estimators, a learning rate of 0.1, default `num_leaves=31`, and `class_weight='balanced'`. LightGBM was the only gradient boosting model trained with explicit class balancing, which proved critical to its performance on the minority class.

4. MODEL EVALUATION

4.1 Evaluation Framework

All models were evaluated on a held-out test set comprising 20% of the dataset (67,635 observations, stratified by class). Given the 87/13 class imbalance, accuracy alone is an insufficient metric. The primary evaluation metrics are:

- Recall (Hazardous Class): The fraction of actual hazardous asteroids correctly identified. In a safety-critical context, false negatives (missed threats) are the costliest error type.
- Precision (Hazardous Class): The fraction of flagged objects that are genuinely hazardous. Lower precision means more false alarms but not missed threats.
- F1-Score (Hazardous Class): The harmonic mean of precision and recall, balancing both concerns.
- AUC-ROC: The area under the receiver operating characteristic curve, measuring the model's ability to rank hazardous objects above non-hazardous ones across all classification thresholds.
- Macro F1: The unweighted average F1 across both classes, penalizing poor performance on the minority class.

4.2 Result Summary

Model	Accuracy	Precision (Haz.)	Recall (Haz.)	F1 (Haz.)	Macro F1
Dummy Baseline	87.4%	—	0.00	0.00	0.47
Logistic Regression	87.4%	0.49	0.00	0.01	0.47

AdaBoost	87.4%	0.00	0.00	0.00	0.47
XGBoost	89.3%	0.70	0.27	0.39	0.66
Random Forest	91.1%	0.71	0.50	0.59	0.77
Linear SVM (C=100)	72.2%	0.30	0.93	0.46	0.64
RBF SVM (C=100, 20k)	70.0%	0.30	0.99	0.46	0.63
LightGBM (balanced)	74.3%	0.33	0.98	0.49	0.66

Figure 3: A summary table of each model's Accuracy, Precision, Recall, F1, and Macro F1 scores.

4.3 Model by Model Analysis

Logistic Regression and AdaBoost both failed to learn the minority class in any meaningful way, producing results identical to the dummy baseline on hazardous-class recall. This failure stems from the absence of class weighting: both models were overwhelmed by the 87% majority class and learned to ignore the hazardous signal entirely. This outcome serves as a cautionary illustration of why accuracy is a misleading metric under class imbalance.

XGBoost improved on the baseline with a hazardous-class recall of 0.27 and F1 of 0.39. While the overall accuracy of 89.3% is higher than Random Forest's 91.1% on the hazardous class, its recall falls well short; it missed 6,267 of 8,547 hazardous objects in the test set. This reflects the cost of training without class balancing on an imbalanced dataset.

Random Forest with balanced class weights achieved the best overall balance of precision and recall among the tree-based models: 0.71 precision and 0.50 recall on the hazardous class, yielding an F1 of 0.59 and the highest macro F1 of 0.77. Its feature importance output also confirmed that the engineered features contributed meaningfully: `kinetic_energy` and `size_to_distance_ratio` ranked among the top features alongside `absolute_magnitude`.

Linear SVM with class balancing produced a dramatically different trade-off: recall of 0.93 on the hazardous class but precision of only 0.30. This means the model catches 93 of every 100 real hazardous asteroids, but 70% of its positive predictions are false alarms. For a triage application where downstream investigation is available to filter false alarms, this trade-off may be operationally acceptable.

RBF SVM pushed recall even higher to 0.99, missing only 84 of 8,547 hazardous objects in the test set. The cost was a further drop in precision to 0.30 and overall accuracy of 70%. The non-linear RBF boundary was better able to wrap around the hazardous asteroid cluster in feature space, but the deep class overlap prevented meaningful precision improvement. The $O(n^2)$ training complexity required subsampling to 20,000 observations, which may have slightly degraded calibration.

LightGBM with balanced class weights achieved the strongest overall performance profile. Its AUC-ROC of 0.906 and hazardous-class recall of 0.98 make it the most practical choice for a triage tool, combining strong sensitivity with competitive discriminative power across all thresholds.

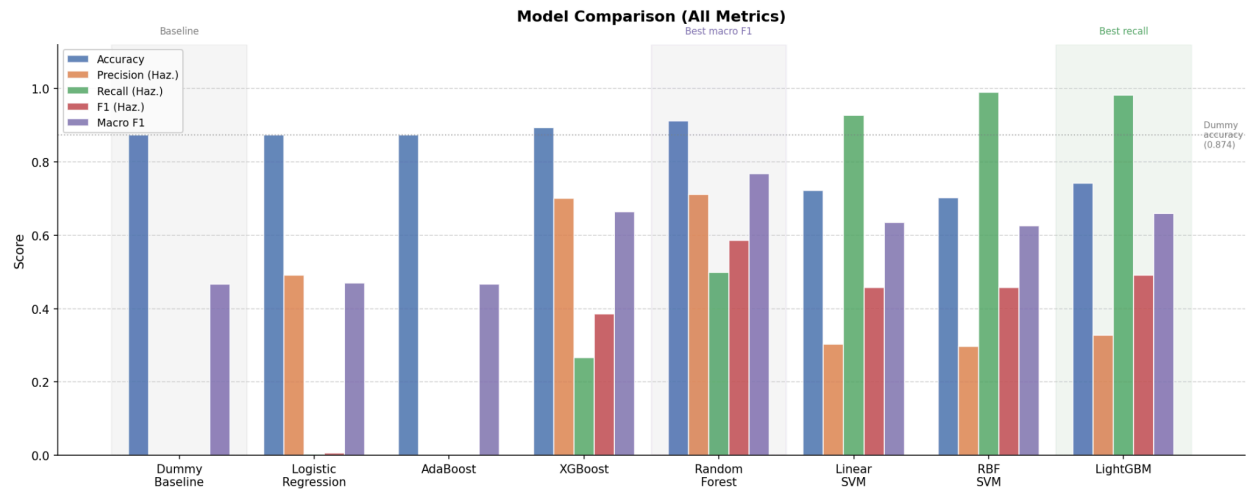


Figure 4: A Bar Graph comparison of each model's performance metrics

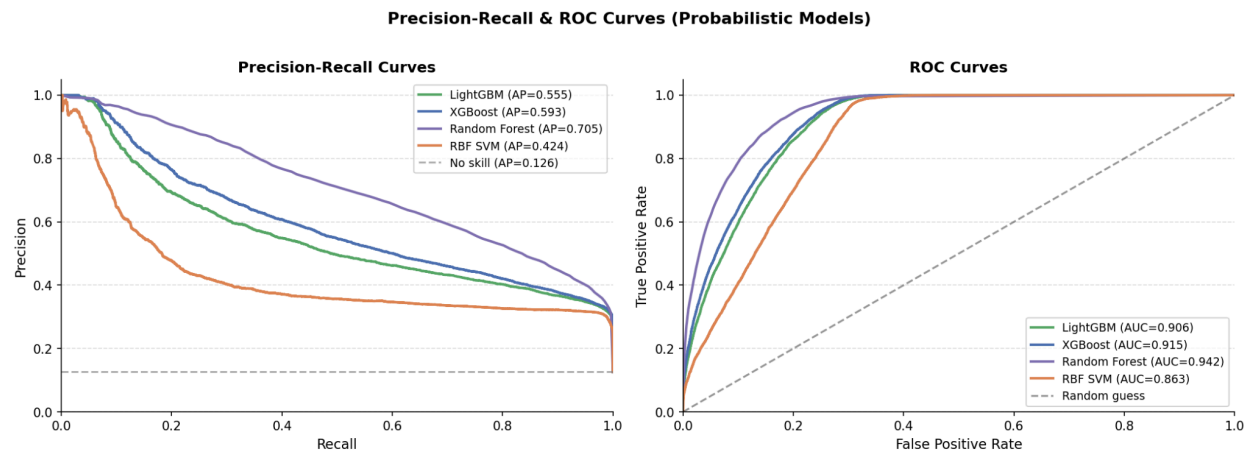


Figure 5: Precision vs Recall curves for the probabilistic models (Left), ROC Curves for the probabilistic models (Right).

4.4 ROC Curve Analysis

The Precision-Recall and ROC curves provide a threshold-independent view of model performance across all four probabilistic classifiers: LightGBM, XGBoost, Random Forest, and RBF SVM. On the ROC curve, Random Forest achieved the highest AUC of (0.942), followed by XGBoost (0.915) and LightGBM (0.906), with RBF SVM trailing at (0.863). The strong performance of Random Forest is notable given that its default threshold metrics suggested it was the more conservative model; the ROC curve reveals it has strong discriminative power across all thresholds, not just at the default 0.5 cutoff. RBF SVM's lower AUC reflects the cost of training on a 20,000-observation subsample rather than the full dataset, which likely degraded its probability calibration.

The Precision-Recall curves are particularly informative given the class imbalance. The no-skill baseline sits at an average precision equal to the positive class proportion (approximately 0.13), and all four models substantially exceed this floor. LightGBM and Random Forest maintain the highest precision at moderate recall levels, making them the most practical choices when downstream investigation capacity is limited. The PR curves also make visible what the summary table obscures: XGBoost's precision collapses rapidly as recall increases, while LightGBM and Random Forest sustain meaningful precision well into high-recall territory.

4.5 Feature Importance

Feature importance rankings from both Random Forest and LightGBM were compared to assess whether the two models agreed on which features drove classification. The results show strong agreement on the top predictors. Absolute magnitude ranked as the dominant feature in both models, consistent with its -0.34 correlation with the target and its physical role as a brightness and size proxy; the same property NASA uses as one of its two official PHA thresholds.

The kinetic energy proxy, an engineered feature combining diameter and velocity, ranked among the top features in both models, validating the physics-motivated design rationale. Size-to-distance ratio and magnitude-velocity interaction also contributed meaningful signal in both models, while miss distance remained the weakest predictor across the board; consistent with its near-zero correlation of -0.007 with the target established in the EDA phase.

The cross-model agreement on feature rankings is an important finding in its own right: it suggests the importance hierarchy reflects genuine signal in the data rather than artefacts of any single model's architecture. The consistent weakness of miss distance across every analysis in this project, correlation, boxplots, PCA loadings, and now feature importance, provides strong evidence that the distance of a given flyby observation carries little information about an asteroid's long-term hazard classification.

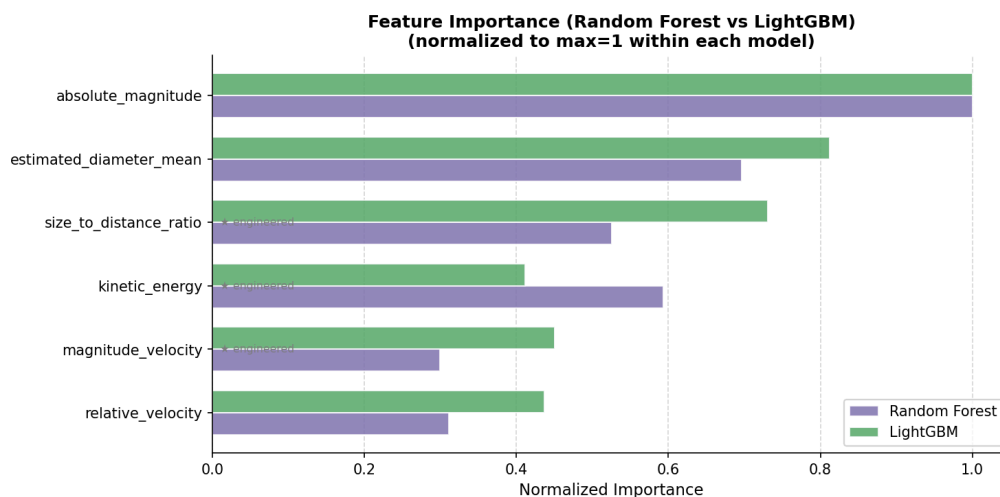


Figure 6: Feature Importance Plot (Random Forest vs LightGBM)

5. RISK SCORE

The risk score converts LightGBM's output probabilities into a continuous 0–100 scale that incorporates both the likelihood of hazard classification and the physical severity of a potential impact. A binary label alone does not fully characterize threat level; two asteroids classified as equally hazardous may differ enormously in size, and therefore in the damage they would cause on impact.

The score is computed as a weighted combination of two components: the LightGBM hazard probability (weighted at 60%) and the asteroid's estimated mean diameter normalized to a 0–1 scale (weighted at 40%). LightGBM was selected as the probability source because it achieved the highest hazardous-class recall (0.98) of all probabilistic models tested, while maintaining a competitive AUC-ROC of 0.906. For a triage tool where missing a genuine threat is the costliest error, recall is the more operationally relevant criterion.

The distribution plot confirms strong class separation at the score level, non-hazardous objects cluster heavily near zero while hazardous asteroids concentrate in the 40–60 range, indicating the combined score provides meaningful separability between classes.

However, inspection of the top 30 highest-scoring asteroids reveals an important limitation: the highest-ranked objects are predominantly non-hazardous false alarms rather than genuinely hazardous asteroids. This occurs because the size weighting can override the probability signal for exceptionally large non-hazardous asteroids; a very large object receives a high severity score even when LightGBM assigns it a low hazard probability. This reflects a genuine tension in risk scoring: physical size and hazard classification are related but not equivalent, and weighting size too heavily can surface large but non-threatening objects above smaller but genuinely dangerous ones. Future work should explore lower alpha values or a nonlinear severity weighting to better balance these two components.

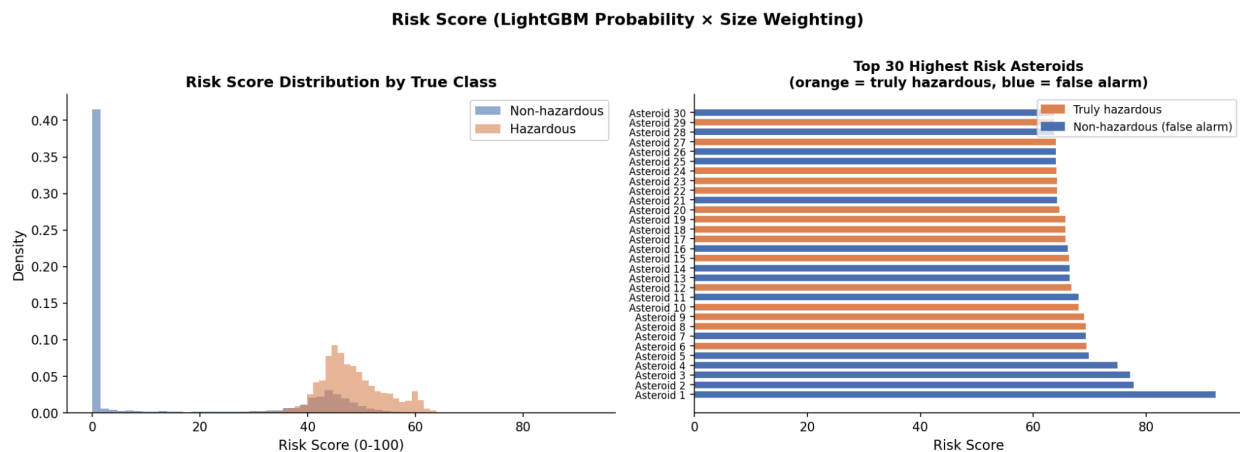


Figure 7: (Left) Risk Score Distribution and (Right) Top 30 Highest Risk Asteroids

6. CONCLUSIONS

6.1 Summary of Findings

This project successfully demonstrated that observable physical and orbital properties of near-Earth asteroids carry meaningful signal for predicting NASA hazard classifications. The best-performing model in terms of recall was LightGBM with balanced class weights, which achieved an AUC-ROC of 0.906 and a hazardous-class recall of 0.98, substantially exceeding all baseline comparisons. Random Forest achieved the highest AUC-ROC overall at 0.942, confirming that tree-based models with explicit class balancing extract the strongest discriminative signal from this feature set.

Several key insights emerged from the analysis. Absolute magnitude is the single most predictive individual feature across both Random Forest and LightGBM importance rankings, consistent with its physical interpretation as a brightness and size proxy and its role as a component of NASA's official PHA threshold. Miss distance proved to be a very poor predictor of hazard status, as the distance of a given flyby observation tells us little about whether an asteroid is classified as hazardous over its full orbital lifetime, a finding that held consistent across every stage of the analysis from correlation to PCA loadings

to feature importance. Physics-motivated feature engineering added genuine predictive value, with the kinetic energy proxy ranking as the second most important feature in both Random Forest and LightGBM, validating the approach of incorporating domain knowledge into feature construction.

Class balancing proved non-negotiable, as models trained without it (Logistic Regression, XGBoost, AdaBoost) failed entirely to learn the minority class despite reporting high overall accuracy, serving as a cautionary illustration of why accuracy is a misleading metric under class imbalance. Finally, no single model dominated across all metrics: Random Forest offered the highest AUC-ROC, LightGBM provided the best recall for a triage application, and RBF SVM maximized sensitivity when false alarms are acceptable.

6.2 Hypothesis Assessment

The project hypothesis, that observable asteroid properties can reliably predict hazard classification, is confirmed with important caveats. The models demonstrate clear, learnable signal, but perfect classification is not achievable without MOID, the missing feature that anchors NASA's actual decision rule. This limitation was identified early, appropriately framed, and does not invalidate the project's contribution as a triage model. The risk score extension further strengthens this framing by producing a ranked priority list that combines model confidence with physical impact severity. However, the risk score analysis also revealed a genuine tension in the scoring design: at $\alpha=0.6$, a small number of exceptionally large non-hazardous asteroids receive scores high enough to dominate the top of the priority list, overriding the probability signal due to their extreme diameter. This finding is consistent with known challenges in real-world planetary defence risk scoring and points toward nonlinear severity weighting as a meaningful area for future refinement.

6.3 Limitations

The primary limitation is the absence of MOID from the dataset, which places a ceiling on achievable classification performance regardless of model architecture. A secondary limitation is that the dataset contains observation-level rows, meaning a single asteroid contributing many close-approach observations could disproportionately influence model training, and future work should aggregate features at the asteroid level using the unique `neo_id` field before modelling. The RBF SVM was trained on a 20,000-observation subsample due to computational constraints, which likely contributed to its lower AUC-ROC of 0.863 relative to the full-dataset models. The risk score's linear size weighting was shown to produce unintuitive rankings at the extremes of the diameter distribution, where size can override the hazard probability signal entirely.

6.4 Next Steps

Several extensions would meaningfully improve the project. Threshold optimization, selecting a classification threshold based on the precision-recall curve rather than defaulting to 0.5, would allow practitioners to explicitly target a desired recall level with maximum achievable precision, which is more operationally useful than the default binary output. Asteroid-level aggregation, collapsing the dataset to one row per unique asteroid using summary statistics such as mean velocity and minimum miss distance, would remove the repeated-measurement confounds present in the current observation-level data. External MOID data from the NASA JPL Small-Body Database could be joined using the `neo_id` field, however this would likely reduce the problem to a deterministic rule rather than a machine learning challenge, as NASA's hazard classification is directly defined by MOID thresholds. A more valuable direction would be to investigate whether MOID itself can be predicted from the observable features in this dataset, effectively using machine learning to estimate the missing orbital parameter rather than to predict hazard classification directly.

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